Toward a four-dimensional Yang–Mills construction

Notebook 14: nonflat backgrounds and energy stability

Research working document

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Abstract

We pass from fixed-reference nonlinear averages to the actual moving-reference averaging operators in a small global connection chart. An explicit endpoint term corrects the change of derivative coordinates. An exact cumulative gauge relation controls ordinary reference backgrounds uniformly through the blocking depth, giving small perturbations of both source averaging operators in physical weighted norms. A separate Wilson plaquette calculation and a form resolvent argument then control a gauge-fixed inverse, including the scalar compression that defines its gauge projection. The inverse estimates concern the energy blocks, with at most one derivative on each side. They apply on a small global physical C1 ball, not yet on the source's full local-frame and retained-history domains.

Status. Research proofs checked through separate internal derivations and audits. No external certification or novelty claim is made. The explicit source-coordinate matches below refer to the deposited transcriptions at the pinned commit, not to newly recovered original CMP98 or CMP99b papers. The action and transpose normalizations are stated explicitly at the point where the Wilson operator is used.

1 The exact change of reference in the averaging differential

The derivative of the corrected average with fixed reference one is not the same coordinate operator as the source average $Q(V)$ based at a nonflat reference V. We now compute their relation, including the endpoint term. The block ratio L is fixed, all logarithms use their common identity branch, and all Hilbert norms use physical volume. This calculation is internally audited at the level of the pinned deposited transcriptions; no fresh verification of the original CMP98 typesetting is claimed.

1.1 Two references and an exact group identity

Write M for the ordinary group average. Raw I:46965–46970 defines

$$\Phi^{\mathfrak{g}}(V,B;y,x)=\frac{1}{i}\log[(e^{iB}V)\langle y,x\rangle V\langle y,x\rangle^{-1}],\qquad \Phi^{\flat}(V,B;c)=\frac{1}{i}\log[M(e^{iB}V)(c)M(V)(c)^{-1}].$$

For U near V set

$$g_y^V(U)=\exp\left(i\sum_{x\in B(y)}L^{-d}\frac{1}{i}\log[U\langle y,x\rangle V\langle y,x\rangle^{-1}]\right). \tag{1}$$

The bridge I:47076–47087, expressly incorporating CMP98 (120), identifies the corrected group increment as

$$C_V(U)(c)=g_{c_-}^V(U)^{-1}M(U)(c)g_{c_+}^V(U)M(V)(c)^{-1}. \tag{2}$$

It equals one at $U = V$. With the logarithmic/linear split I:24344–24349, the same bridge gives

$$C_V(e^{iB}V) = e^{i\mathcal{Q}(V,B)}, \qquad \mathcal{Q}(V,B) = LQ(V)B + O(B^2). \tag{3}$$

This is the deposited identification of the middle coordinate; the earlier transcription does not itself reproduce CMP98 (120). We use the explicit bridge, rather than guessing that omitted input formula.

Set $a_y(U) = g_y^1(U)$. The corrected map with fixed identity reference is

$$C_1(U)(c) = a_{c_-}(U)^{-1} M(U)(c) a_{c_+}(U). \tag{4}$$

For fixed V define $k_y(U;V) = g_y^V(U)^{-1} a_y(U) a_y(V)^{-1}$. Direct substitution and cancellation, preserving the factor order, give

$$C_1(U)(c)C_1(V)(c)^{-1} = \mathrm{Ad}_{a_{c_-}(V)^{-1}}\left[k_{c_-}(U;V)^{-1} C_V(U)(c)\, \mathrm{Ad}_{M(V)(c)}\, k_{c_+}(U;V)\right]. \tag{5}$$

In particular no multiplicative cocycle for the endpoint mean has been assumed.

For $U(s) = e^{isB}V$ let

$$\eta_y^1(V;B) = \frac{1}{i}\frac{d}{ds}\Big|_0 a_y(U(s))a_y(V)^{-1}, \qquad \eta_y^V(B) = \frac{1}{i}\frac{d}{ds}\Big|_0 g_y^V(U(s)), \qquad Z_V B(y) = \eta_y^1(V;B) - \eta_y^V(B).$$

Define the *unscaled* endpoint difference by

$$d_W z(c) = z(c_-) - \mathrm{Ad}_{W(c)}\, z(c_+). \tag{6}$$

This explicit definition fixes the sign independently of other covariant-derivative conventions. All middle factors in (5) equal one at $s = 0$, so differentiating gives

$$\frac{1}{i}\frac{d}{ds}\Big|_0 C_1(e^{isB}V)(c)C_1(V)(c)^{-1} = \mathrm{Ad}_{a_{c_-}(V)^{-1}}[LQ(V)B(c) - d_{M(V)}(Z_V B)(c)]. \tag{7}$$

As a separate check, I:46977 defines $LQ^\flat(V)B = \Phi_1^\flat(V;B)$. Differentiating the general bridge directly at $B = 0$ gives

$$LQ(V)B = LQ^\flat(V)B - d_{M(V)}\overline{r}(V;B), \qquad \overline{r}(V;B)(y) = \sum_x L^{-d}\Phi_1^{\mathfrak{g}}(V;B;y,x). \tag{8}$$

Thus the later rooted operator $Q^\flat$ and the corrected source operator Q are distinct. Substitution into (7) reproduces the direct derivative of (4), confirming its endpoint sign.

1.2 An explicit formula for the endpoint correction

For a small complex Lie-algebra element X put

$$\mathcal{J}_X K = \int_0^1 \mathrm{Ad}_{e^{itX}}\, K\, dt, \qquad \frac{1}{i}\frac{d}{ds}\Big|_0 e^{i(X+sK)}e^{-iX} = \mathcal{J}_X K. \tag{9}$$

The integral power series and a Neumann series show that $\mathcal{J}_X$ and its inverse are analytic near zero, and $\|\mathcal{J}_X - I\| + \left\|\mathcal{J}_X^{-1} - I\right\| \le C\,\|X\|$. This does not require unitarity.

Write

$$V\langle y,x\rangle = e^{iY_x}, \qquad \overline{Y}_y = \sum_x L^{-d} Y_x, \qquad r_x(V;B) = \Phi_1^{\mathfrak{g}}(V;B;y,x).$$

The last quantity includes the actual contour orientations, transports and repetitions. Applying (9) to each contour logarithm and then to the exponential of their mean proves

$$Z_V B(y) = \mathcal{J}_{\overline{Y}_y} \sum_x L^{-d} \mathcal{J}_{Y_x}^{-1} r_x(V;B) - \sum_x L^{-d} r_x(V;B). \tag{10}$$

In particular $Z_1 = 0$. The leading background term is

$$Z_V B = \frac{i}{2}\left([\overline{Y}, \bar{r}] - \sum_x L^{-d}[Y_x, r_x]\right) + O((\max_x |Y_x|)^2 \max_x |r_x|).$$

There is no general identity forcing this difference of means of commutators to vanish. The endpoint correction in (7) must therefore be retained.

1.3 The physical derivative conversion and its bound

At fine spacing h write $V = e^{ihA}$ and define the fixed-reference physical map by $C_1(e^{ihA})(c) = e^{iLh\mathcal{F}_h(A)(c)}$. For an ordinary potential variation K, the fine angular left increment is $h\mathcal{J}_{hA}K$ and the coarse left increment is $Lh\mathcal{J}_{Lh\mathcal{F}_h(A)}D\mathcal{F}_h(A)K$. Substituting these into (7) gives

$$\mathcal{J}_{Lh\mathcal{F}_h(A)}D\mathcal{F}_h(A)K = \mathrm{Ad}_{a_-(V)^{-1}}\left[Q_h(V)\mathcal{J}_{hA}K - L^{-1}d_{M(V)}Z_V(\mathcal{J}_{hA}K)\right]. \tag{11}$$

Consequently the exact operator conversion is

$$\boxed{Q_h(V) = \mathrm{Ad}_{a_-(V)}\,\mathcal{J}_{Lh\mathcal{F}_h(A)}D\mathcal{F}_h(A)\mathcal{J}_{hA}^{-1} + L^{-1}d_{M(V)}Z_V.} \tag{12}$$

Here $Q_h(V)$ is the normalized source coefficient placed between the physical spaces at h and Lh. Neither $d_{M(V)}$ nor Z_V contains an inverse mesh factor. The argument of Z_V before division is $h\mathcal{J}_{hA}K$; its linearity cancels h against the coarse Lh and leaves exactly $1/L$.

Proposition 1.1 (Actual one-step source comparison)**.** *For fixed L, if $h\|A\|_\infty$ lies in a sufficiently small common complex neighborhood, then*

$$\left\|Q_h(e^{ihA}) - Q_h(1)\right\|_{p,h\to p,Lh} \le C_{L,p}h\|A\|_\infty, \qquad p = 2,\infty. \tag{13}$$

The constants are independent of h, refinement and compatible periodic volume. The same estimate holds for absolute coefficient row sums and physical-adjoint row sums, after changing the fixed constant.

Proof. Each one-step contour has bounded length at fixed L. Products of possibly nonunitary complex transports therefore give $\max_x |Y_x| + |\overline{Y}| \le C_L h\|A\|_\infty$ and $\|r_x(V;\cdot)\|_{\infty\to\mathrm{color}} \le C_L$. Equation (10) implies $\|Z_V\|_{\infty\to\infty} \le C_L h\|A\|_\infty$.

For the physical Hilbert bound expand Z_V into its finitely many one-step bond coefficients. Each is analytic, vanishes at $V = 1$, and is bounded by $C_L h\|A\|_\infty$. Rows and columns have fixed finite incidence; the physical adjoint supplies the fixed volume ratio L^d. The Schur bound thus gives the same order from fine physical L^2 to coarse-site physical L^2, with no macroscopic path-trace estimate. The endpoint map $d_{M(V)}$ has two entries per row, fixed column incidence and bounded complex adjoints, so it is uniformly bounded in both norms.

The other coordinate factors in (12) differ from the identity by at most $C_L h\|A\|_\infty$. Combine this with the proved one-step estimate $\|D\mathcal{F}_h(A) - Q_h(1)\|_{p\to p} \le D_p h\|A\|_\infty$ and the flat contractions to obtain (13). Equivalently, the entire corrected source coefficient in (3) is a finite local analytic function of the link logarithms hA, equal to $Q_h(1)$ at zero. Its coefficientwise Taylor bound and the same finite incidence prove the row and physical-adjoint row assertions directly. Formula (12) records the coordinate factors hidden by that shorter proof. Periodic multiplicity counting preserves all constants. □

For a multilevel source comparison this estimate must be evaluated at the actual ordinary reference backgrounds $M^t(U)$, whose amplitudes are controlled in the following argument. It does not license replacing them by the corrected fixed-reference iterates. It also proves no nonflat inverse or minimizing-field theorem by itself.

2 Ordinary reference backgrounds and their averaging operators

The corrected iteration at reference 1 and the ordinary group average have different flat linearizations. We relate the complete configurations by an explicit gauge, then estimate the actual moving-reference source operators at the ordinary backgrounds. This argument does not equate those operators with the ordinary derivative of the fixed-reference corrected iteration.

Fix odd $L \geq 3$, $M = L^j$, $\eta = M^{-1}$ and

$$h_t = L^t \eta \quad (0 \leq t \leq j), \qquad \sum_{s<t} h_s = \frac{h_t - \eta}{L-1}. \tag{14}$$

We use compatible periodic lattices or the full lattice, globally defined complex Lie-algebra bond fields, and physical volume in every Hilbert space. All constants below are independent of j, M and compatible periodic volume. They may depend on fixed L, dimension, color norm and the averaging prescription. Matrix estimates use a submultiplicative norm; fixed norm equivalences can be absorbed in the constants.

2.1 The exact endpoint identity

Let M denote the ordinary group-valued average. For a configuration V near the identity put

$$\alpha(V;y) = L^{-d} \sum_{x \in B(y)} \frac{1}{i} \log V(\Gamma_{y,x}), \qquad a(V;y) = e^{i\alpha(V;y)}. \tag{15}$$

The corrected group map at reference 1 is

$$\mathsf{C}(V)(c) = a(V;c_-)^{-1} \mathsf{M}(V)(c) a(V;c_+). \tag{16}$$

This is the specialization of deposited Raw I:24313–24325, equivalently its explicit coordinate bridge I:47082–47086. For a general reference the endpoint factor also carries an adjoint by the average of that reference. It disappears here because the reference is 1.

Use the gauge convention $V^g(c) = g(c_-) V(c) g(c_+)^{-1}$. Ordinary gauge covariance is

$$\mathsf{M}(V^g) = \mathsf{M}(V)^{g|_{\mathrm{coarse}}}. \tag{17}$$

The deposit records restriction to base points at I:43018–43026, and expressly invokes complexified gauge covariance from CMP98 (11) at I:82901–82904. In the common identity neighborhood it also follows from similarity covariance of the analytic products, logarithms and means. All gauges used below stay inside that neighborhood.

These are source matches to the pinned deposited transcription. The full original CMP98 paper has not been independently reverified in this calculation. In particular, the bridge's explicit incorporation of its omitted coordinate (120) is kept as the source convention, rather than replaced by a guessed coordinate formula.

2.2 Uniform ordinary connection amplitudes

Let $U = e^{i\eta A}$ and $a = \|A\|_\infty$. Write

$$C_0 = U, \quad C_{t+1} = \mathsf{C}(C_t), \qquad V_t = \mathsf{M}^t(U). \tag{18}$$

The fixed-reference theorem of Notebook 13 gives a radius $\rho > 0$, physical corrected fields A_t and a constant D_∞, such that for $a < \rho$

$$C_t = e^{ih_t A_t}, \qquad \|A_t\|_\infty \le 2a, \qquad A_{t+1} = \mathcal{F}_{h_t}(A_t), \tag{19}$$

and

$$\|D\mathcal{F}_h(B)\|_{\infty\to\infty} \le 1 + D_\infty h \|B\|_\infty . \tag{20}$$

The derivative bound is used on a common smaller ball, which contains the corrected trajectories and their straight-line interpolations with translates.

Theorem 2.1. *There are $a_* > 0$ and $C_V < \infty$, independent of refinement and volume, such that if $\|A\|_\infty < a_*$, every ordinary reference V_t has a holomorphic physical logarithm*

$$B_t = (ih_t)^{-1} \log V_t, \qquad \|B_t\|_\infty \le C_V \|A\|_\infty . \tag{21}$$

Moreover there is an explicit vertex gauge g_t with $V_t = C_t^{g_t}$ and $g_t^{\pm 1} - 1 = O(h_t \|A\|_\infty)$. All these maps read a bounded final-scale neighborhood.

Proof. Let m_Γ be the largest number of oriented bonds in a one-step contour. It is fixed when L is fixed; the conservative bound $m_\Gamma \le dL$ suffices. Decrease a_* so that $2m_\Gamma a_* \le \log(3/2)$. By (19),

$$\|C_t(\Gamma) - 1\| \le e^{2m_\Gamma h_t a} - 1 \le \tfrac{1}{2}.$$

The logarithm series and the bound $e^s - 1 \le 3s/2$ for $0 \le s \le \log(3/2)$ give, with slack,

$$\|\alpha(C_t; y)\| \le K_\Gamma h_t a, \qquad K_\Gamma = 8m_\Gamma. \tag{22}$$

Set

$$g_0 = 1, \qquad g_{t+1}(y) = g_t(y) a(C_t; y), \qquad y \in \Lambda_{t+1}. \tag{23}$$

The lattices are nested, so the restriction of g_t is defined. If $V_t = C_t^{g_t}$, then (17) and (16) imply

$$V_{t+1}(c) = g_t(c_-) a(C_t; c_-) C_{t+1}(c) a(C_t; c_+)^{-1} g_t(c_+)^{-1}.$$

Thus $V_t = C_t^{g_t}$ by induction, with exactly the product order in (23).

The ordered exponential products and (14)–(22) give

$$\left\| g_t^{\pm 1} \right\| \le \exp\left(\frac{K_\Gamma a (h_t - \eta)}{L - 1} \right), \tag{24}$$

$$\left\| g_t^{\pm 1} - 1 \right\| \le \exp\left(\frac{K_\Gamma a (h_t - \eta)}{L - 1} \right) - 1. \tag{25}$$

The second bound follows by telescoping; no unitarity or commutativity is required. Put

$$K_V = 2 + \frac{2K_\Gamma}{L - 1}, \qquad C_V = 3K_V. \tag{26}$$

The same product estimate for $V_t(c) = g_t(c_-) e^{ih_t A_t(c)} g_t(c_+)^{-1}$ yields

$$\|V_t(c) - 1\| \le e^{K_V a h_t} - 1. \tag{27}$$

Take also $K_V a_* \le \log(3/2)$. The analytic logarithm then satisfies

$$\|\log V_t(c)\| \le 2(e^{K_V a h_t} - 1) \le 3K_V a h_t,$$

which proves (21).

Further reduce a_*, if necessary, to place the right side of (27) in the fixed one-step analytic neighborhood of M. This also justifies the construction inductively before each ordinary average is applied. Holomorphy follows from the fixed-reference theorem and the explicit products and logarithms. Each one-step dependency has physical radius at most a fixed multiple of h_t, so the accumulated radius is bounded by a multiple of the geometric sum (14). Alternatively, use the deposited two-final-block support of the corrected bond iterate. □

2.3 Physical first differences and curvature

Supremum smallness alone does not give bounded physical curvature. In a commuting direction take A_2 alternating between a and $-a$ at neighboring sites in direction 1, with all other components zero. The plaquette angle is of order ηa, hence the curvature divided by η^2 is of order a/η. This obstruction is already present at $t = 0$.

Theorem 2.2. *In addition to Theorem 2.1, assume*

$$b = \max_\nu \|\nabla_{\eta,\nu} A\|_\infty < \infty. \tag{28}$$

Then, with fixed constants,

$$\max_\nu \|\nabla_{h_t,\nu} A_t\|_\infty \le e^{2D_\infty a/(L-1)} b, \tag{29}$$

$$\max_\nu \|\nabla_{h_t,\nu} B_t\|_\infty \le C e^{Ca} b, \tag{30}$$

$$h_t^{-2} \|V_t(\partial p) - 1\| \le C e^{Ca}(b + a^2). \tag{31}$$

These estimates hold for real or complex fields on compatible periodic lattices, with the same constants.

Proof. The fixed-block map $\mathcal{F}_h$ commutes with translations by Lh in each coordinate direction. For such a translation T, integrate its derivative along the segment from B to TB. This segment remains in the same supremum ball. Equation (20) gives

$$\|\mathcal{F}_h(TB) - \mathcal{F}_h(B)\|_\infty \le (1 + D_\infty h \|B\|_\infty) \|TB - B\|_\infty .$$

Telescoping L fine steps shows $\|TB - B\|_\infty / (Lh) \le \|\nabla_{h,\nu} B\|_\infty$. Iterating, using (19) and (14), proves (29). This requires only coarse translation covariance, not invariance under an arbitrary fine translation.

On the same fixed coordinate ball, differentiation of the finite contour formula gives

$$\left\|D_A \alpha(e^{ihA})\right\|_{\infty\to\infty} + \left\|D_A a(e^{ihA})\right\|_{\infty\to\infty} \le C_\Gamma h. \tag{32}$$

Indeed, each differentiated bond exponential contributes h; the other factors have bounded product norm. The logarithm derivative is bounded on $\|V - 1\| \le 1/2$, as is the final exponential derivative. No fine derivative is taken here.

For $s < t$, translation by h_t preserves the scale-s block construction. By (32), telescoping h_t/h_s input steps and using (29),

$$\|a(C_s; y + h_t e_\nu) - a(C_s; y)\| \le C h_s h_t e^{2D_\infty a/(L-1)} b. \tag{33}$$

The inverse factors obey the same estimate. Telescope the ordered products defining g_t at the two vertices, bound the undifferentiated products by (24), and sum h_s. This gives

$$\left\| g_t(y + h_t e_\nu)^{\pm 1} - g_t(y)^{\pm 1} \right\| \le C e^{Ca} h_t^2 b. \tag{34}$$

The corresponding difference of $C_t(c) = e^{i h_t A_t(c)}$ is bounded by the same expression. The identity $V_t = C_t^{g_t}$ therefore bounds the difference of V_t across $h_t e_\nu$ by $C e^{Ca} h_t^2 b$. The logarithm is uniformly Lipschitz in the half-unit ball, by integrating its derivative on the convex segment. Dividing the logarithm difference by h_t^2 proves (30). The case $t = 0$ is the input estimate.

For curvature, first work with C_t. Pairing the two parallel sides of a plaquette, the difference formula for exponentials bounds the opposite-side differences by $C h_t^2 \max_\nu \|\nabla_{h_t,\nu} A_t\|_\infty$. The remaining constant-side product is a group commutator, bounded by $C h_t^2 a^2$. Thus

$$\|C_t(\partial p) - 1\| \le C h_t^2 \left(e^{2D_\infty a/(L-1)} b + a^2 \right).$$

The ordinary plaquette is conjugated by g_t at its initial vertex. Equation (24) proves (31). For real fields the gauges are unitary, so a unitarily invariant norm avoids this conjugation loss. Periodization preserves the translation identities and all telescoping estimates, including on short compatible tori. □

2.4 Actual bond averages in physical norms

Let $Q_h(V)$ be the normalized linear coefficient of the corrected relative increment at reference V, held fixed. It is the one-step source operator of CMP98 (124), transcribed at Raw I:24355–24369. The explicit bridge I:47082–47086 expresses it through finite local analytic operations. At $V = 1$ it is the straight-string operator Q_h^{st}.

For an operator from spacing h to spacing H, define the absolute physical Schur norm by

$$\|P\|_{\mathrm{av}} = \max \left\{ \sup_c \sum_b \|P_{cb}\|, \quad \sup_b (H/h)^d \sum_c \|P_{cb}\| \right\}. \tag{35}$$

This controls the bounded-field norm, the bounded-field norm of the physical adjoint and the physical L^2 norm. It is submultiplicative across scales: the volume ratios multiply. For the flat normalized straight-string map both row sums equal one. In fact each input bond occurs L times among the L^d translated strings of length L, so its ordinary column sum is L^{-d} and its physical adjoint column sum is one.

Finite-dimensional local analyticity and fixed-L row and column incidence give constants $r_Q, C_Q > 0$ with

$$\left\| Q_h(e^{iX}) - Q_h^{\mathrm{st}} \right\|_{\mathrm{av}} \le C_Q \|X\|_\infty, \qquad \|X\|_\infty < r_Q. \tag{36}$$

To verify this directly, differentiate each local coefficient on a smaller closed complex polydisc and sum its finitely many incidences. The physical column factor is L^d, independent of h. The separate exact change-of-reference calculation identifies the logarithm Jacobians and endpoint correction that enter these coefficients; this shorter norm argument uses their already explicit local analytic definition.

The actual source product uses the ordinary backgrounds:

$$Q_j(U) = Q_{h_{j-1}}(V_{j-1}) \cdots Q_\eta(V_0). \tag{37}$$

This is the composition recorded at Raw I:25147–25155 and I:47046–47061. By Theorem 2.1, $\|\log V_t\|_\infty \le C_V a h_t$. Reduce a_* so that $C_V a_* < r_Q$. Submultiplicativity and an exact telescoping of

the difference of products prove

$$\|Q_j(U)\|_{\mathrm{av}} \le \exp\left(\frac{C_Q C_V a}{L-1}\right), \tag{38}$$

$$\left\|Q_j(U) - Q_M^{\mathrm{st}}\right\|_{\mathrm{av}} \le \exp\left(\frac{C_Q C_V a}{L-1}\right) - 1. \tag{39}$$

For a partial product ending at h_t, replace $(L-1)^{-1}$ in the exponent by $(h_t-\eta)/(L-1)$. These are actual source operator estimates, obtained without renaming $D\mathcal{F}^{(j)}(A)$ as $Q_j(U)$.

2.5 Scalar averages and finite physical range

The scalar average called Q'_j in the deposit, and Q_0 in the homogeneous flat analysis, is

$$(Q'_j(U)\lambda)(y) = L^{-jd} \sum_{x \in B^j(y)} \mathrm{Ad}_{U(\Gamma^{(j)}_{y,x})}\, \lambda(x). \tag{40}$$

Raw I:23913–23920 states this formula. Its contour follows successive block centers. The description at I:17798–17810 uses level-s ordinary averaged configurations on the legs, with at most dL bonds per leg. The ordinary amplitude bound and (14) therefore imply

$$\left\|U(\Gamma^{(j)}_{y,x})^{\pm 1}\right\| \le e^{Cah_j}, \qquad \left\|\mathrm{Ad}_{U(\Gamma^{(j)}_{y,x})} - I\right\| \le e^{Cah_j} - 1. \tag{41}$$

At a fixed level the blocks partition the fine vertices. Each input scalar vertex occurs in one row with weight L^{-jd}, canceled by the physical adjoint volume factor. Consequently

$$\left\|Q'_j(U)\right\|_{\mathrm{av}} \le e^{Cah_j}, \qquad \left\|Q'_j(U) - Q'_j(1)\right\|_{\mathrm{av}} \le e^{Cah_j} - 1. \tag{42}$$

This scalar operator is distinct from the bond average and from the symbol Q_0 used for the main term of CMP98 (125).

Partition the spaces into compatible physical cells of side one, with their physical Hilbert measures. The finite multilevel reading radius is bounded independently of refinement. For bond averages the deposit states the sharper two-final-block support at I:23927–23930 and I:24332–24336. If R bounds the physical reading radius and N_R the number of possible neighboring cells, every cell block has operator norm at most the global Hilbert norm. Hence

$$\|P\|_{\kappa,\mathrm{cell}} := \max\left\{\sup_x \sum_y e^{\kappa d(x,y)} \|P_{xy}\|_{2\to 2}\,, \sup_y \sum_x e^{\kappa d(x,y)} \|P_{xy}\|_{2\to 2}\right\} \le N_R e^{\kappa R} \|P\|_{2\to 2} \tag{43}$$

for such a finite-range P. Applying this to the differences in (39) and (42) gives uniform $O(a)$ bounds in the physical-cell weighted Schur norm. Periodic distance and fixed cover multiplicities give the same result on small compatible tori.

2.6 Local holomorphy and scope

All constructions above are holomorphic on the same complex ball $\|A\|_\infty < a_*$. At finite regulators this follows from products of finite-dimensional holomorphic maps. Their uniform bounds and bounded physical reading ranges also give the corresponding bounded-field local maps on the full lattice. In particular $A \mapsto Q_j(e^{i\eta A})$ and $A \mapsto Q'_j(e^{i\eta A})$ are holomorphic in the weighted operator norm just

defined, with a uniform bound on every smaller ball. For example, if that bound is C_0 and an argument lies at distance at least r from the boundary of the chosen ball, the multivariable Cauchy formula gives

$$\|D^n Q_j(A)[K_1,\dots,K_n]\|_{\kappa,\mathrm{cell}} \le C_0 n! \left(\frac{e}{r}\right)^n \prod_{m=1}^{n} \|K_m\|_\infty , \tag{44}$$

and the same statement for Q'_j. One obtains it using polydisc radii $r/(n\,\|K_m\|_\infty)$ and $n^n \le e^n n!$. Directions of zero norm are handled by multilinearity. This is a bounded-background derivative statement; no Hilbert norm is substituted for a background direction without another estimate.

The result supplies uniform ordinary reference trajectories, their physical C^1 and curvature bounds when the initial first differences are bounded, and actual source averaging operators close to their flat counterparts. It concerns a globally trivialized homogeneous class. The next section uses these estimates to control an explicitly normalized Wilson Hessian and its moving projected gauge inverse. The minimizing field is constructed separately in Notebook 15 on a small global physical $C^{1,\alpha}$ ball; Notebook 16 controls its nonlinear local responses and classical coarse action. Mixed retained-history domains, strict background-support expansions and an interacting Yang–Mills ultraviolet limit remain separate. The original-primary-paper verification limitation stated in Section 2.1 remains in force.

3 A nonflat Wilson Hessian and its full energy inverse

The averaging bounds above close the remaining averaging hypotheses of an energy-form perturbation argument. We first calculate the Wilson Hessian, including its normalization. Its bare action, physical pairing and left-exponential chart match the deposited conventions in dimension four. The resulting theorem uses the deposited homogeneous averaging maps; the graded-curl and effective Hessian identifications remain separate, as stated below.

3.1 Action, charts, and an exact local decomposition

Write $\eta = M^{-1} \le 1$ and $\tau = N_c^{-1}\,\mathrm{Tr}$. Fine fields have physical pairing $\langle B, C\rangle_\eta = \eta^d \sum_e \tau(B_e C_e)$. The sum is over positive bonds; plaquettes below are counted once, with $\mu < \nu$. On Hermitian fields this is the real Hilbert pairing; its complex extension is bilinear. Superscript T always denotes that transpose, with no conjugation of a complex background. Define

$$\mathcal{S}_\eta(U) = \eta^{d-4} \sum_p \left(1 - \frac{1}{2}\tau(P_p + P_p^{-1})\right), \qquad U_e(s,t) = e^{i\eta(sB_e + tC_e)} U_e, \tag{45}$$

and define Δ_U by the mixed derivative of this action. There is no coupling prefactor in (45). At $U = 1$, $\Delta_1 = d^T d$, where d is the ordinary physical plaquette curl.

Let $U_e = e^{i\eta a_e}$, with

$$\epsilon_0 = \|a\|_\infty , \qquad \epsilon_1 = \|\nabla a\|_\infty , \qquad F = \epsilon_1 + \epsilon_0^2. \tag{46}$$

The field can be complex; ϵ_0 is bounded by a fixed small constant. Comparing opposite plaquette sides by the integral formula for matrix exponentials, and bounding the residual group commutator, gives

$$\|P_p - I\| + \left\|P_p^{-1} - I\right\| \le C\eta^2 F. \tag{47}$$

Indeed opposite potentials differ by at most $\eta\epsilon_1$, so their exponentials differ by $C\eta^2\epsilon_1$. For the remaining two constant potentials the commutator costs $C\eta^2\epsilon_0^2$. All path transports involved here have a fixed number of links and norm at most $e^{C\eta\epsilon_0}$.

For the loop $e_1, \dots, e_4$ around p, set $V_i = U_{e_1} \cdots U_{e_{i-1}}$. A negative traversal of the positive bond f has exact chart increment $B_{-f} = -\operatorname{Ad}_{U_f^{-1}} B_f$. Put $X_i^B = \operatorname{Ad}_{V_i} B_{e_i}$ and

$$\mathcal{C}_U B = \eta^{-1} \sum_i X_i^B. \tag{48}$$

The exact plaquette variation is $P_p(s,t) = \prod_i e^{i\eta(sX_i^B + tX_i^C)} P_p$. Every transport difference from identity is $O(\eta\epsilon_0)$, and therefore

$$\mathcal{C}_U = d + E_U, \qquad \|E_U\|_{\infty\to\infty} + \left\|E_U^T\right\|_{\infty\to\infty} \le C\epsilon_0. \tag{49}$$

The transpose norm uses physical measure. Fixed plaquette incidence also gives the same estimate in a weighted physical-cell Schur norm.

Writing $S_B = \sum_i X_i^B$, direct mixed differentiation gives

$$\partial_s \partial_t P_p|_0 = -\eta^2 \left\{ \frac{S_B S_C + S_C S_B}{2} + \frac{1}{2} \sum_{i<j} ([X_i^B, X_j^C] + [X_i^C, X_j^B]) \right\} P_p. \tag{50}$$

For the oriented density $-\tau(P_p)$ the principal contribution is $\eta^d \tau(((\mathcal{C}_U B)(\mathcal{C}_U C) + (\mathcal{C}_U C)(\mathcal{C}_U B))P_p)/2$. The remaining contribution is exactly

$$\frac{\eta^d}{2\eta^2} \sum_{i<j} \tau\left(([X_i^B, X_j^C] + [X_i^C, X_j^B])(P_p - I) \right), \tag{51}$$

because a commutator has trace zero. The reverse loop has curl $-\operatorname{Ad}_{P_p^{-1}} \mathcal{C}_U$; cyclicity removes this adjoint transport from its principal quadratic term. Thus

$$\Delta_U = \mathcal{C}_U^T M_U \mathcal{C}_U + V_U, \qquad \tau(X M_{U,p} Y) = \frac{1}{4}\tau((XY + YX)(P_p + P_p^{-1})), \tag{52}$$

where V_U is half (51) plus half its reverse-loop version. Projection onto the fixed traceless Lie space is implicit in the definition of M_U. These coefficients are bilinearly symmetric and holomorphic. Equation (47) implies

$$\|M_U - I\| \le C\eta^2 F, \qquad \|V_U\|_{\infty\to\infty} + \left\|V_U^T\right\|_{\infty\to\infty} \le CF. \tag{53}$$

Each curl has row and transpose-row norm $O(\eta^{-1})$. Consequently the ordinary bounded local operator $W_U = \Delta_U - \mathcal{C}_U^T \mathcal{C}_U$ has row and transpose-row norm CF: the two inverse mesh factors are paid for by η^2 in (53). In particular

$$\Delta_U - d^T d = d^T E_U + E_U^T d + E_U^T E_U + W_U = S_v^T V_{\mathrm{kin}} S_v, \quad S_v = (I, \nabla), \tag{54}$$

with $\|V_{\mathrm{kin}}\|_\kappa \le C(\epsilon_0 + \epsilon_1 + \epsilon_0^2)$. Here ∇ contains all physical fine differences of bond components; the curl is a fixed component combination of them. The scalar gauge derivative is separately

$$D_U \lambda(e) = \eta^{-1}(\operatorname{Ad}_{U_e} \lambda(e_+) - \lambda(e_-)) = D\lambda + E_s \lambda, \qquad \|E_s\|_\kappa \le C\epsilon_0. \tag{55}$$

All coefficients in these two estimates have fixed fine range.

3.2 The scalar compression and a form-resolvent lemma

For physical-cell block operators use the Banach algebra norm

$$\|T\|_\kappa = \max\left\{\sup_x \sum_y e^{\kappa d(x,y)}\,\|T_{xy}\|\,, \sup_y \sum_x e^{\kappa d(x,y)}\,\|T_{xy}\|\right\}. \tag{56}$$

The norm is submultiplicative and invariant under transpose. Fine-cell spaces have measure η^d, so it introduces no power of the number of fine sites. The earlier flat inverse theorem gives

$$\left\|S_v G_0 S_v^T\right\|_\kappa \le K_v, \quad G_0 = (d^T d + D R_0 D^T + Q^T a_{\mathrm{pen}} Q)^{-1}, \tag{57}$$

for a_{pen} in a fixed compact positive interval. Both flat averages $Q = Q_M^{\mathrm{st}}$ and Q_0, and their transposes, have uniform weighted norms by physical row/column contraction and fixed final-scale range.

For clarity the additional scalar input is derived explicitly. Let $G_0' = (\mathcal{L} + Q_0^T a' Q_0)^{-1}$, $\mathcal{L} = D^T D$, where a' lies in a fixed compact positive interval, and let $S_s = (I, D)$. In coarse Bloch aliases write P_r for the fine Laplacian multiplier, a_r for the scalar average, and

$$\begin{aligned} A &= a_0 a_0^-, & R_{s,k} &= \sum_{r\neq 0} \frac{a_r a_r^-}{P_r^k}, \\ D_{s,2} &= A + P_0^2 R_{s,2}, & N &= P_0 + a' A + a' P_0 R_{s,1}. \end{aligned} \tag{58}$$

The scalar Woodbury formula has central entry $(R_{s,1} + (a')^{-1})/[A + P_0(R_{s,1} + (a')^{-1})]$, mixed entry $-a_0^- a_s/[P_s(A + P_0(R_{s,1} + (a')^{-1}))]$, and a high diagonal P_r^{-1} plus a rank-one correction. These canceled expressions, high-alias gaps and alias Parseval give $\left\|S_s G_0' S_s^T\right\|_\kappa \le K_s$. They also give the exact compression formula

$$C_0 = Q_0 (G_0')^2 Q_0^T = \frac{D_{s,2}}{N^2}, \qquad C_0^{-1} = \frac{N^2}{D_{s,2}}. \tag{59}$$

On an enlarged real Bloch cube, $A \ge c > 0$ and the high remainders are nonnegative and uniformly bounded. Thus $N, D_{s,2}$ are bounded above and away from zero. Their uniformly bounded holomorphic preliminary tube can be shrunk by a fixed amount to retain these lower bounds. Periodic physical-cell reconstruction and contour shifting prove $\|C_0\|_\kappa + \left\|C_0^{-1}\right\|_\kappa \le K_c$. The zero fiber gives $C_0 = (a')^{-2}$; for $M = 1$ it gives $(P_0 + a')^{-2}$. No inverse of the unpenalized zero mode is used.

We will repeatedly use the following elementary form identity. If $G = \mathcal{A}^{-1}$, $K = SGS^T$ has weighted norm at most K_*, and I is a component of S, then for $K_* \|V\|_\kappa < 1$,

$$(\mathcal{A} + S^T V S)^{-1} = G - G S^T V (I + KV)^{-1} S G, \tag{60}$$

and the augmented block has norm at most $K_*/(1 - K_* \|V\|_\kappa)$ and differs from K by at most $K_*^2 \|V\|_\kappa / (1 - K_* \|V\|_\kappa)$. The Neumann series in (56) proves these assertions and holomorphy. Multiplication verifies the inverse at each fixed spacing on the finite or infinite lattice. If the real baseline $\mathcal{A}$ is positive definite, the same smallness gives relative-form coercivity for real symmetric perturbations.

3.3 Conditional closure of the projected gauge inverse

Theorem 3.1. *Suppose, in one common weighted algebra, that*

$$\begin{gathered} D_U = D + E_s, \quad Q_s(U) = Q_0 + F_s, \quad Q_v(U) = Q + F_v, \quad \Delta_U = d^T d + S_v^T V_{\mathrm{kin}} S_v, \\ \|E_s\|_\kappa + \|F_s\|_\kappa + \|F_v\|_\kappa + \|V_{\mathrm{kin}}\|_\kappa \le \varepsilon. \end{gathered} \tag{61}$$

Define

$$\begin{aligned} G'_U &= (D_U^T D_U + Q_s(U)^T a' Q_s(U))^{-1}, \\ C_U &= Q_s(U)(G'_U)^2 Q_s(U)^T, \\ R_U &= I - G'_U Q_s(U)^T C_U^{-1} Q_s(U) G'_U, \\ \mathcal{A}_U &= \Delta_U + D_U R_U D_U^T + Q_v(U)^T a_{\mathrm{pen}} Q_v(U). \end{aligned} \tag{62}$$

For a sufficiently small uniform $\varepsilon_* > 0$, if $\varepsilon \le \varepsilon_*$ all these inverses exist and

$$\left\| S_s G'_U S_s^T \right\|_\kappa + \left\| C_U^{-1} \right\|_\kappa + \left\| R_U \right\|_\kappa + \left\| S_v \mathcal{A}_U^{-1} S_v^T \right\|_\kappa \le C. \tag{63}$$

Each displayed block differs from its flat counterpart by at most $C\varepsilon$. Constants are uniform in refinement and compatible periodic volume; the full-lattice statement also holds. Holomorphic coefficient families give holomorphic inverse families. On real data R_U is the orthogonal projection off $\operatorname{ran}(G'_U Q_s(U)^T)$.

Proof. The scalar precision difference is

$$D^T E_s + E_s^T D + E_s^T E_s + Q_0^T a' F_s + F_s^T a' Q_0 + F_s^T a' F_s.$$

It equals $S_s^T V_s S_s$ with $\|V_s\|_\kappa \le C_s \varepsilon$. The form identity controls the scalar energy block and its $O(\varepsilon)$ difference when $K_s C_s \varepsilon \le 1/2$. Product expansion gives $\|C_U - C_0\|_\kappa \le C'_s \varepsilon$. Equation (59) and

$$C_U^{-1} = (I + C_0^{-1}(C_U - C_0))^{-1} C_0^{-1}$$

then control its inverse if $K_c C'_s \varepsilon \le 1/2$. The formula for R_U gives its bound and its $O(\varepsilon)$ difference. Its projection assertion is the Gram projection formula; for complex coefficients it is a bilinear holomorphic continuation rather than a positivity claim.

The gauge difference is exactly

$$D_U R_U D_U^T - D R_0 D^T = D(R_U - R_0) D^T + E_s R_U D^T + D R_U E_s^T + E_s R_U E_s^T. \tag{64}$$

The divergence D^T on bonds is a fixed shifted component combination of ∇. Keeping these derivatives in S_v, (64) is $S_v^T V_g S_v$ with norm $C_g \varepsilon$. The average-penalty difference $Q^T a_{\mathrm{pen}} F_v + F_v^T a_{\mathrm{pen}} Q + F_v^T a_{\mathrm{pen}} F_v$ is a bounded I, I coefficient of size $C_a \varepsilon$. Thus $\mathcal{A}_U = \mathcal{A}_0 + S_v^T V_{\mathrm{tot}} S_v$, with $\|V_{\mathrm{tot}}\|_\kappa \le C_v \varepsilon$. A second application of (60), with $K_v C_v \varepsilon \le 1/2$, proves the result. □

The theorem controls G_U, ∇G_U, $G_U \nabla^T$ and $\nabla G_U \nabla^T$, with exponential coarse-cell locality. It does not propagate one-sided second derivatives by this argument: differentiating $G_0 S_v^T$ twice would require an unavailable total third-order baseline block.

3.4 Penalty independence and the homogeneous Wilson conclusion

The scalar penalty is unnecessary for identifying the projection. Fix $\mathcal{L}_U = D_U^T D_U$ and Q_s, and let $G_b = (\mathcal{L}_U + Q_s^T b Q_s)^{-1}$, $E_b = G_b Q_s^T$. Whenever the inverse and Gram inverse exist,

$$\operatorname{ran} E_b = \{ v : \mathcal{L}_U v \in \operatorname{ran} Q_s^T \}. \tag{65}$$

One inclusion follows by applying $\mathcal{L}_U$ to $G_b Q_s^T$. For the other, $\mathcal{L}_U v = Q_s^T w$ implies $v = G_b Q_s^T (w + b Q_s v)$. The range is therefore independent of the symmetric coarse penalty b. Its Gram inverse makes it nondegenerate, and $E_b (E_b^T E_b)^{-1} E_b^T$ is the unique bilinear projection onto this range along its orthogonal complement. Thus R_U is penalty independent. The identity holds directly for complex data whenever the stated inverses exist, and on real data it is the ordinary orthogonal projection identity.

We may consequently construct R_U using $a' = 1$. It agrees with a differently penalized source formula wherever that formula is defined, without estimating an unprinted scalar penalty sequence. An estimate for that other scalar inverse G'_U itself still requires its own normalization bounds.

Corollary 3.2. *Fix odd $L \geq 3$, $M = L^j$, and the homogeneous full or periodic lattice. Let $Q_v(U)$ be the deposited product of corrected one-step derivatives at the ordinary reference trajectory, as in (37), and let $Q_s(U)$ be the deposited scalar transported average (40). Define Δ_U by (45), D_U by (55), and R_U by (62) with auxiliary scalar penalty one. For a_{pen} in a fixed compact positive interval, there are $\rho, \kappa, C > 0$, independent of j and compatible volume, such that*

$$\|a\|_\infty + \|\nabla a\|_\infty < \rho \quad \Longrightarrow \quad \left\| (I, \nabla)(\Delta_U + D_U R_U D_U^T + Q_v(U)^T a_{\mathrm{pen}} Q_v(U))^{-1} (I, \nabla)^T \right\|_\kappa \leq C. \tag{66}$$

The inverse is holomorphic on this complex global C^1 ball, and its augmented block differs from the flat block by at most $C(\|a\|_\infty + \|\nabla a\|_\infty)$.

Proof. Equations (39), (42), and (43) give $\|Q_v(U) - Q\|_\kappa + \|Q_s(U) - Q_0\|_\kappa \leq C\epsilon_0$. Their mechanism is the ordinary-reference estimate $\|\log V_t\| \leq C h_t \epsilon_0$, a physical one-step coefficient difference $O(h_t \epsilon_0)$, and the geometric sum $\sum_{t<j} h_t \leq 1/(L-1)$; no derivative of a different nonlinear composition is substituted. Equations (54) and (55) supply the remaining hypotheses of Theorem 3.1. □

The bare source normalization is matched at the deposited-transcription level. Raw I lines 453–467 normalize $\operatorname{tr} 1 = 1$, count unoriented plaquettes once and define $A(U) = \sum_p (1 - \operatorname{Re} \operatorname{tr} P_p)$; lines 472–481 place the coupling separately. Thus this bare action is exactly (45) in dimension four. Lines 12756–12760 give the physical trace pairings with fine measure η^d and their bilinear transposes. Lines 15471–15478 and 15643–15648 define Δ as the quadratic coefficient in the simultaneous left chart $\mathsf{A}(e^{i\eta A} U_0)$, exactly the chart in (45). Lines 43103–43107 again specify the bare Wilson action for constrained minimization. These passages resolve the action-prefactor, physical-sum and chart conventions used here. Lines 13583–13593 define the full source operator $\Delta + D_U R D_U^T + Q^T a Q$, with $a = 1$ at unit scale. At a nonzero global potential A_0, the ordinary-coordinate Hessian of $A \mapsto \mathcal{S}_\eta(e^{i\eta A})$ also contains the background current paired with the second derivative of the change to this left chart. Matching the action does not remove that term or identify the two Hessian matrices. The separate source claim about $\Delta - D_U^T D_U$ still requires matching its precise graded one-form covariant curl, and no mixed retained-history effective Hessian is identified with this bare homogeneous Hessian. These checks concern the pinned deposit, not a fresh verification of the original journal proofs.

Corollary 3.2 is a theorem for its explicitly defined Wilson operator and deposited homogeneous averages. It does not establish nonflat pointwise weak norms, nonflat current domains, mixed retained histories, arbitrary nontrivial reference connections, or an interacting continuum construction.

References

[1] Pinned Yang–Mills-LLM-experiment account, commit `e392ef45013a581664bab76b908671 70b30fe10e`. The exact locations and remaining source-identification checks are stated alongside the corresponding arguments above.

[2] Companion research proofs: Notebook 09, the full flat fine inverse and scalar gauge identities; Notebook 12, pointwise and Schauder regularity on the flat current domain; Notebook 13, the complete fixed-reference nonlinear averaging ball and constraint chart.